Rajiv Gandhi Institute of Petroleum Technology, Jais Amethi

Department of Computer Science and Engineering

Mid-Semester Examination AY 2025-26 (ODD Sem)

Subject: Object-Oriented Methodologies **Course Number: CS224**

Date & Time: 17-09-2024, 03:00 PM to 05:00 PM **Total Marks:** 30

INSTRUCTIONS

I. **This paper contains a total of six questions, with the marks for each question indicated alongside it. Students are making any necessary assumptions as required for clarification.**

---

1. Define a C++ class **BankAccount** with private data members: accountNumber, holderName, and balance. **[5]**

(a) Write public member functions: Deposit(), Withdraw(), DisplayDetails(), and Transfer(BankAccount, double) (Write the bodies of the above functions from your side)

(b) In main() function, create an array of pointers to BankAccount objects for 5 customers.

(c) Use the pointers to perform a transaction where ₹2000 is deducted from one customer's account and added to another customer's account by calling the Transfer() function.

(d) Finally, display all account details using pointer access.

2. Write a C++ program that repeatedly asks the user to enter a positive integer and then prints the factorial of that integer.

(a) The program should have a factorial function that will compute the factorial of the input received by the user. **[2]**

(b) Apply a suitable constraint that will make sure that if any user enters zero or any negative number, then the program terminates and exits from the main() **[1]**

(c) Furthermore, after showing the factorial value, create one more function that asks the user whether they want to continue, and if the user presses the 'Yes' option, then find the square root of the factorial value. However, if the user presses the 'No' option, then find the square of the factorial value. **[2]**

3. Write a C++ program that creates three classes: Complex1, Complex2, and ComplexSum.

(a) Class Complex1 stores a complex number. **[1]**

(b) Class Complex2 stores another complex number. **[1]**

(c) Receive the input values of the data members of both classes using mutator functions. **[1]**

(d) Class ComplexSum compute the sum of both complex numbers and store inside it. **[1]**

(e) Use an accessor function to display the result of ComplexSum class. **[1]**

4. (a) Define Abstraction in Object-Oriented Programming. **[1]**

(b) Differentiate between pointer and reference with a suitable example. **[1]**

(c) Explain the benefits of Encapsulation in real-world applications with example. **[1]**

(d) Why are variables in C++ required to have a data type at the time of declaration? **[1]**

(e) Which OOP feature allows software systems to be designed in a way that changing one part of the system has minimal impact on others? **[1]**

5. Find the output of the following questions. Also, give a proper justification for your answer. If there is any compile time error, then mentation the reason of that error. **[5]**

(a) #include <iostream>
using namespace std;
void update (int &a, int &b)
{
a = a + b;
b = a - b;
a = a - b;
}
int& choose (int &x, int &y, bool flag)
{
return (flag ? x : y);
}
int main()
{
int p = 10, q = 20, r = 30;
update(p, q);
cout << p << q << endl;
choose(p, r, true) = 100;
cout << p << q <<r<< endl;
choose(q, r, false) += 50;
cout << p << q <<r<< endl;
return 0;
}
	(b) #include<iostream>
	using namespace std;
	class sum
	{
	int x, y, total;
	void display ()
	{
	total = x + y;
	cout <<total;
	}
	public:
	sum (int a, int b)
	{
	x=a; y=b*2;
	}
	};
	int main ()
	{
	sum s (9,5);
	s. display ();
	return 0;
	}
		(c)#include<iostream>
		using namespace std;
		class bita
		{
		static int c;
		public:
		static void count()
		{
		c = c++ * ++c-(--c)*c++;
		}
		static void display()
		{
		count();
		cout << "\nValue of c" << c;
		}
		};
		int bita::c = 20;
		int main() {
		bita::count();
		bita::display();
		return 0;
		}

6. Find the correct answer of the following questions. Also, give a valid reason of your answer.

(a) Which of the following is a key characteristic of structured programming? **[1]**

(I) Use of classes and objects
(II) Division of program into small functions or modules
(III) Use of polymorphism
(IV) All the Above

(b) Which OOP feature allows different functions with the same name to behave differently depending on the context? **[1]**

(I) Encapsulation
(II) Polymorphism
(III) Inheritance
(IV) Abstraction

(c) Which of the following is not a C++ token? **[1]**

(I) Identifier
(II) Statement
(III) Keyword
(IV) Constant

(d) Which of the following best describes an Abstract Data Type (ADT)? **[1]**

(I) A predefined C++ data type like int or char
(II) A user-defined type with implementation details hidden
(III) A library function
(IV) A pointer type

(e) Which of the following is true about the structure of a C++ program? **[1]**

(I) The main() function can be written before including header files.
(II) Only one main() function can exist in a program.
(III) namespace std must always be explicitly declared.
(IV) Execution can start from any function, not necessarily main().

****************************End of Question Paper *************************